PROBLEM6:

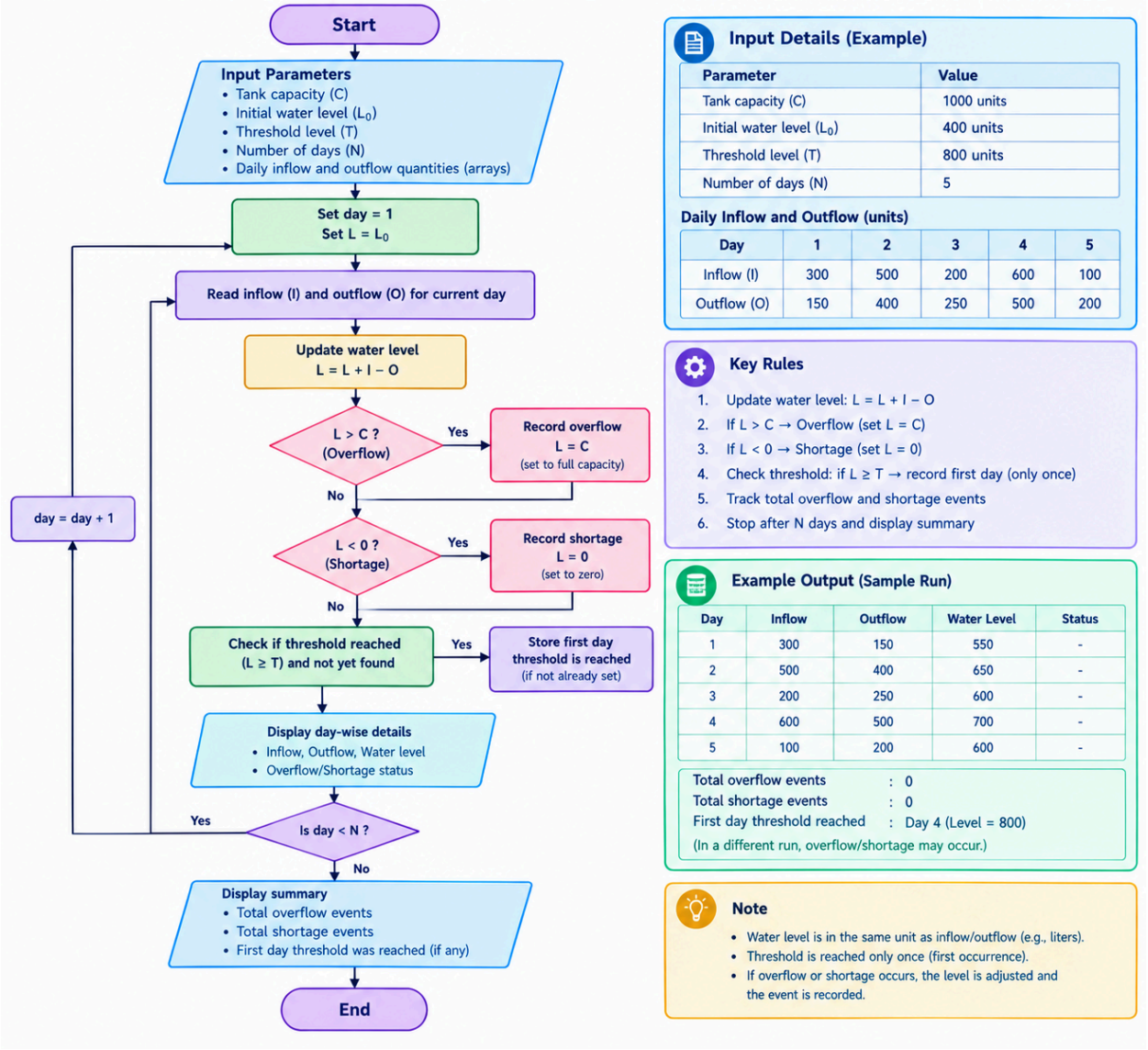
PROBLEM STATEMENT:

Multi-Day Water Tank Simulation: Process changing inflow and outflow quantities, update water levels, detect overflow and shortages, and determine the first interval in which the tank reaches a specified threshold.

ALGORITHM:

1. Start.
2. Input the tank capacity, initial water level, threshold level, and number of days.
3. Validate the input values.
4. For each day:
    - Input the water inflow and outflow quantities.
    - Calculate the tentative water level:
      \[
      \text{Level}=\text{Previous Level}+\text{Inflow}-\text{Outflow}
      \]
    - If the tentative level exceeds tank capacity, record the overflow and set the final level to the tank capacity.
    - If the tentative level is below zero, record the shortage and set the final level to zero.
    - Otherwise, update the water level normally.
    - Check whether the threshold has been reached for the first time.
    - Display the day's water level, overflow, shortage, and threshold status.
5. If the threshold was never reached, display an appropriate message.
6. Stop.

FLOWCHART:



PSEUDOCODE:

BEGIN

INPUT capacity, initial_level, threshold, days

level = initial_level

```
first_interval = 0

FOR day = 1 TO days
    INPUT inflow, outflow

    tentative_level = level + inflow - outflow
    overflow = 0
    shortage = 0

    IF tentative_level > capacity THEN
        overflow = tentative_level - capacity
        level = capacity
    ELSE IF tentative_level < 0 THEN
        shortage = -tentative_level
        level = 0
    ELSE
        level = tentative_level
    ENDIF

    IF level >= threshold AND first_interval = 0 THEN
        first_interval = day
    ENDIF

    DISPLAY day, level, overflow, shortage
```

```
    ENDFOR

    IF first_interval = 0 THEN
        DISPLAY "Threshold not reached"
    ELSE
        DISPLAY "First threshold interval:", first_interval
    ENDIF
END
```

PROGRAM:

```
#include <stdio.h>

int main() {
    int days, day, first_interval = 0;
    double capacity, level, threshold;
    double inflow, outflow, tentative;
    double overflow, shortage;

    printf("Enter tank capacity: ");
    scanf("%lf", &capacity);

    printf("Enter initial water level: ");
    scanf("%lf", &level);
```

```
printf("Enter threshold level: ");
scanf("%lf", &threshold);

printf("Enter number of days: ");
scanf("%d", &days);

if (capacity <= 0 || level < 0 ||
    level > capacity || threshold < 0 ||
    threshold > capacity || days <= 0) {
    printf("Invalid input values.\n");
    return 1;
}

for (day = 1; day <= days; day++) {
    printf("\nDay %d\n", day);

    printf("Enter inflow: ");
    scanf("%lf", &inflow);

    printf("Enter outflow: ");
    scanf("%lf", &outflow);

    if (inflow < 0 || outflow < 0) {
        printf("Invalid flow quantities.\n");
```

```
    return 1;
}

tentative = level + inflow - outflow;
overflow = 0;
shortage = 0;

if (tentative > capacity) {
    overflow = tentative - capacity;
    level = capacity;
} else if (tentative < 0) {
    shortage = -tentative;
    level = 0;
} else {
    level = tentative;
}

if (level >= threshold && first_interval == 0) {
    first_interval = day;
}

printf("Water level: %.2f\n", level);
printf("Overflow: %.2f\n", overflow);
printf("Shortage: %.2f\n", shortage);
```

```
    }

    if (first_interval == 0) {
        printf("\nThreshold was not reached.\n");
    } else {
        printf("\nFirst threshold reached on day %d.\n",
                first_interval);
    }

    return 0;
}
```

OUTPUT:

```
Enter tank capacity: 100
Enter initial water level: 60
Enter threshold level: 80
Enter number of days: 4

Day 1
Enter inflow: 20
Enter outflow: 10
Water level: 70.00
Overflow: 0.00
Shortage: 0.00
```

Day 2

Enter inflow: 40

Enter outflow: 5

Water level: 100.00

Overflow: 5.00

Shortage: 0.00

Day 3

Enter inflow: 0

Enter outflow: 30

Water level: 70.00

Overflow: 0.00

Shortage: 0.00